
EE/CprE/SE 492 WEEKLY REPORT 6

October 30, 2025 - November 13, 2025

Group number: 16

Project title: Outdoor Roomba

Client &/Advisor: Prof. Diane Rover

Team Members/Role:

Brodie M Bates - Embedded system

Daniel Vergara Pinilla - Machine Learning - Backend

Mitchell D Owen - Project Tester

Om Shukla - Frontend Developer

Sudipta Halder - Simulation developer

○ Weekly Summary

This week we created a retractable for the power shovel. Designed and printed the brackets for the wheels of the encoders. Got multiple encoders working on a multiplex for the I2C. Ordered Metal and plexiglass to build the final version of the robot.

We made some changes to the final project. We have removed the weather-proofing and durability requirements. Mainly because of our lack of expertise in the field of mechanical engineering.

○ **Past week's accomplishments**

- Sudipta Halder: Made smaller beams, so the bottle can go through more easily. Implemented free rtos on esp32. Got the shovel to be printed and tested, redesigned the shovel retraction mechanism. Started work on SLAM on the Raspberry Pi.
- Brodie Bates: Worked on the motor encoders. Wrote code to output the degree of the motor when it's running.
- Daniel Pinilla: Worked on Multiple target detection and decision-making for that process
- Om Shukla: Added connections for create account and robot endpoints.
- Mitchell Owen: Reassembling our project with new parts and measuring what parts still need to be cut out. Also reevaluating some of the components that still need to be attached to the robot.

- **Pending issues**

- Mitchell Owen: We are falling behind with how little time we have left. We need to focus on our key issues and get a final project set up to show to the board. We still have enough time, but we need to focus and get it done.

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u>	<u>Hours these two weeks</u>	<u>HOURS cumulative</u>
Brodie Bates	I wrote code to read the motor encoders. Looked at the design document for potential changes.	10	67
Om Shukla	Added connections for create account and robot endpoints.	10	46
Sudipta Halder	Redesigning the brackets, implementation of free rtos on esp32 and pico2. Started work on SLAM on the raspberry pi	38	121
Daniel Mauricio Vergara	Changes in Backend, deployed new version of website, and worked on new logic for multiple objects detection	22	60
Mitchell Owen	Assembling the new brackets on the robot. Measuring the new base and side panels. Reorganizing the casing of the robot to better secure the motors. Drafting and organizing the wiring to cleanly fit in a storage box located in the base of the robot.	25	70

○ **Plans for the upcoming week**

- Sudipta Halder: Implementation of the server. Build the final version of the robot, and conduct user testing over thanksgiving break.
- Brodie Bates: This week I will be working on finishing the encoders as well as helping assemble the robot and get permanent mounting spots for our electronics.
- Om Shukla: Since the server has been updated with the new design of the website, now I can test the login endpoints and debug them.
- Mitchell Owen: Finish the assembly of the robot with the new components. Test the robot to make sure it still works and put on finishing touches to get it prepared for presenting.
- Daniel Pinilla: Testing for all components together as well as new servo for the motor

- **Summary of weekly advisor meeting**

The first week's advisor meeting focused on the quality of our work and areas we need to improve for our presentation and design document. Our advisor meeting presentation required significant improvement, particularly in terms of formatting and some of the information presented.

In the second week, we discussed what everyone is working on and how some aspects of our project might need to wait until we have something to demonstrate. Some of these parts are the encoders, which are nice to have but ultimately not needed for the robot to move, as well as some of the SLAM parts. We should, however, mention that we have worked on it and get a video for proof.